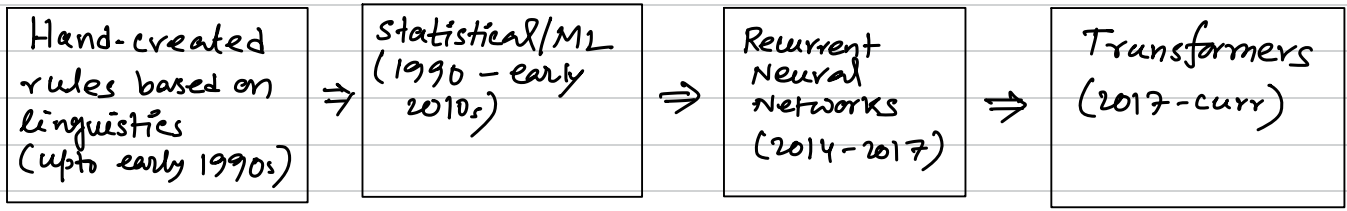


NLP

[illegible]

How to get a new input sentence ready to be "fed" into a DNN

- What is the best way to "feed" this 8×100 table of numbers to a DNN
- What if we "aggregate" the vectors?
 - Sum of vectors — Count encoding
 - "OR" the vectors — Multi Hot encoding
- This aggregation approach is called the Bag of words model.

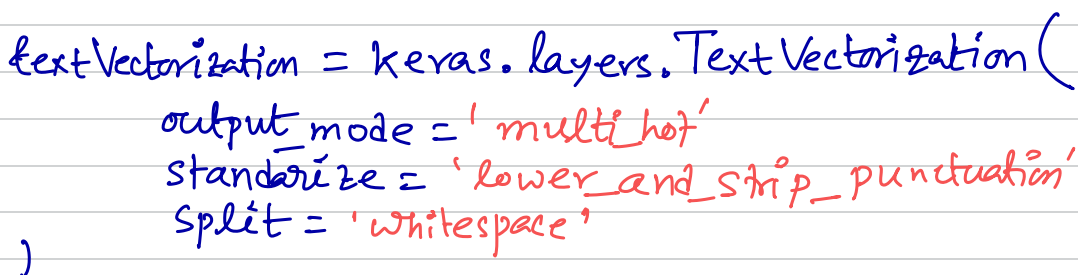
UNK a ... an ... car ... cat ... hat ... mat ... ox ... on ... sat ... ten ... the ... za

count encoding

Multihot encoding

⇓ we lost the order of the words
is it cat sat on mat?
OR, mat sat on cat?

ST/E $\rightarrow$ Standardize | Tokenize | Index | Encode



```
dataset = [  
    "I write, erase, rewrite",  
    "Erase again, and then"  
    "A poppy blooms",  
    "Hola! are you laid-back in Mexico"  
]
```

```
textVectorization.adapt(dataset)  
vocabulary = textVectorization.getVocabulary()  
len(vocabulary) #17  
print(vocabulary) # ['UNK', 'erase', 'you', 'write'...]
```

```
test_sentence = "I write, rewrite, and still rewrite again"  
encoded_sentence = textVectorization(test_sentence)  
print(encoded_sentence)
```